The Joy of Computing using Python

Assignment 3

NOTE: Python 3.7 has been used for this Assignment

1. What is the expected output for the following code?

```
cart=['coffee','sugar','cheese','butter']
for item in cart:
        if item=='sugar':
                print('jaggery')
        else:
                print(item)
```

   (a) ['coffee', 'jaggery', 'cheese ', 'butter ']

   (b) ['coffee ', 'sugar ', 'cheese ', 'butter ']

   (c) coffee
   jaggery
   cheese
   butter

   (d) coffee jaggery cheese butter

   CORRECT ANSWER: (c)

   Solution: The items in the list named 'cart' are only displayed, not the list itself. So the first two options are wrong. The print statement adds by default a newline character at the end. Therefore only option three is right.

2. Which of the following code prints the sum of weights of people in the lift.

   (a)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in weights:
        sum=sum+w
print(sum)
```

   (b)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in range(len(weights)):
        sum=sum+w
print(sum)
```

   (c)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in weights:
sum=sum+w
print(sum)
```

   (d)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in weights:
        sum=w
print(sum)
```

CORRECT ANSWER: (a)

First option is correct. Indenting is wrong in the second and third options whereas in the third option, the sum is not calculated by adding the items in the list in the last option.

3. Consider a python list named 'book_titles'. Pick the statement to add 'Who moved my cheese?' as the third item.

   Given: book_titles = ['Exam Warriors', 'Evil in the Mahabharata', '6 TIMES THINNER', 'The Driver in the Driverless Car', 'Evolution']

   (a) book_titles.append(2,'Who moved my cheese?')
   (b) book_titles.insert(2,'Who moved my cheese?')
   (c) book_titles.insert(3,'Who moved my cheese?')
   (d) book_titles.append(3,'Who moved my cheese?')

   CORRECT ANSWER: (b)

   Solution: book_titles.insert(2,'Who moved my cheese?') is correct.

   insert() function is used to add an item to the list at a desired index position. The indexing starts from 0. Therefore option two is correct.

4. Pick the relevant output for the given code.

```
n=[1 ,4 ,2 ,8 ,21 ,17]
n. reverse ()
print(n)
```

   (a) [1, 2, 4, 8, 17, 21]
   (b) [21, 17, 8, 4, 2, 1]
   (c) [17, 21, 8, 2, 4, 1]
   (d) [1, 4, 2, 8, 21, 17]

   CORRECT ANSWER: (c)

   Solution: [17, 21, 8, 2, 4, 1] is correct. The reverse() function exactly reverses the sequence of elements in the list.

5. Specify the purpose of 'break' statement inside a nested loop.

   (a) Ends execution of the program
   (b) Ends execution of the outermost loop
   (c) Skips the current iteration of the loop
   (d) Ends the execution of the loop

   CORRECT ANSWER: (d)

6. You are given a list, 'marks' scored by 30 students. Identify the instruction to find the 2% trimmed mean for the given data.

   (a) m=stats.trim_mean(marks,0.2)
   (b) m=stats.trim_mean(marks,0.03)
   (c) m=stats.trim_mean(30,0.02)
   (d) m=stats.trim_mean(marks,0.02)

   CORRECT ANSWER: (d)

7. How will you simulate 'Rolling a Dice' with six faces by making use of 'random' library?

   (a) roll= random.choice(1,2,3,4,5,6)
   (b) roll= random.range(1,5)
   (c) roll= random.randint(1,6)

(d) roll= random.random(6)

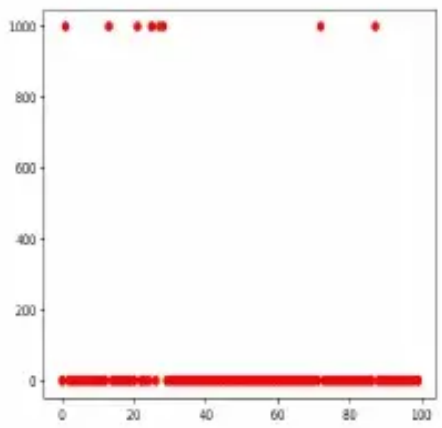
CORRECT ANSWER: (c)

8. Which of the plots in the options is most likely to be generated from the following code?

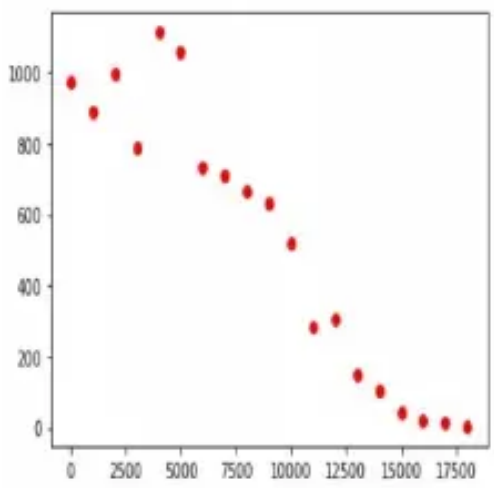
```
import random
import matplotlib.pyplot as plt

def play():
        amt=0
        for i in range(0,100):
                r=random.randint(1,1000)
                if(r!=random.randint(1,1000)):
                        amt=amt
                else:
                        amt=amt+1000
        return amt

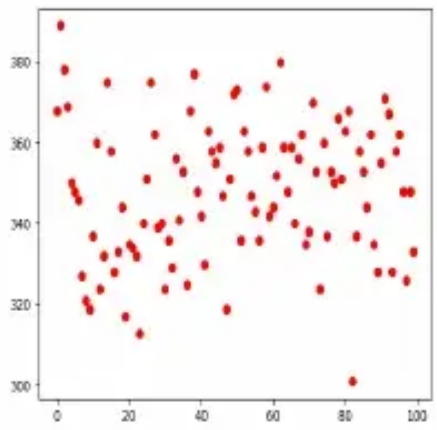
l=[]
for j in range(0,100):
        s=0
        for i in range(0,100):
                s=s+play()
        l.append(s)
x=[]
y=[]
for each in list(set(l)):
        x.append(each)
        y.append(l.count(each))
plt.plot(x,y,'ro')
plt.show()
```



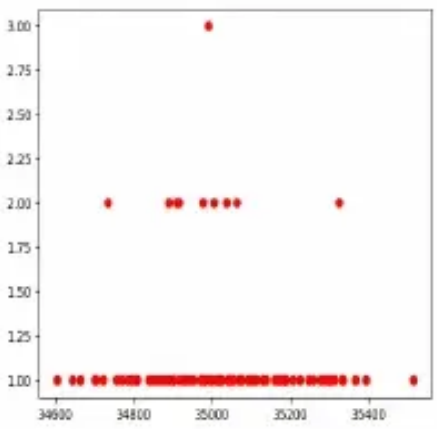
(a)



(b)



(c)



(d)

CORRECT ANSWER: (b)

9. Assuming, there is no file named 'file.txt' on my computer, what does the following code do?

```
with open('file.txt','w') as f:
        f.write('Hey! I am writing.');
f.close()
with open('file.txt','w') as f:
        f.write('Hey I am writing the second line.');
f.close()
with open('file.txt','r') as f:
        print(f.read())
f.close()
```

(a) Shows error

(b) Displays: Hey I am writing the second line

(c) Displays: Hey! I am writing.Hey I am writing the second line.

(d) Displays: Hey! I am writing.

CORRECT ANSWER: (b)

10. Predict the output

```
my_para='i am to go to KT in A'
print(list(my_para))
```

(a) `['i', ' ', 'a', 'm', ' ', 't', 'o', ' ', 'g', 'o', ' ', 't', 'o', ' ', 'K', 'T', ' ', 'i', 'n', ' ', 'A']`

(b) `['i', 'a', 'm', 't', 'o', 'g', 'o', 't', 'o', 'K', 'T', 'i', 'n', 'A']`

(c) `['i', 'am', 'to', 'go', 'to', 'KT', 'in', 'A']`

(d) `['i', ' ', 'am',' ', 'to',' ', 'go',' ', 'to',' ', 'KT',' ', 'in',' ', 'A']`

CORRECT ANSWER: (a)

First option is correct because every element including whitespace in the input string becomes an element in the output list.